

Year-round Vehicle-to-Grid Enabled Frequency Regulation Potential of Electric Bus Public Transport Fleets During Overnight Charging

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Abstract

Frequency regulation is one of the most promising Vehicle-to-Grid (V2G) services that can be provided by electric vehicles, due to short durations requiring minor depth of discharge, and considerable revenues that can be earned per event. In addition to stationary batteries and passenger EVs, electric buses of public transport fleets could be very suitable with their expanding market share, large energy storage, charging/discharging power and scheduled idle periods overnight at depot. This study focuses on frequency regulation service provision potential of electric bus public transport fleets during their overnight charging and idle parking times at depots. A year-round analysis is conducted with the 2024 system frequency data of National System Energy Operator (NESO) of UK, revealing overnight frequency regulation event needs and fleet availabilities, considering commonly preferred electric bus technical specifications, typical route characteristics and V2G-enabled charger capabilities.

1. Introduction

The electric vehicle sector is expanding to include not only passenger cars but also a wide range of vehicle types such as buses, minibuses, vans, trucks, and lorries. In 2023, for the first time in history, annual electric bus sales across Europe surpassed those of all other fuel types [1]. According to the International Energy Agency's (IEA) Global EV Outlook 2023 report, approximately 27 countries have committed to selling only zero-emission buses and trucks by 2040 [2]. The United States and the European Union have proposed stricter emission regulations for heavy-duty vehicles. In addition, more than 60,000 electric buses (E-buses) were sold worldwide in 2022, accounting for approximately 5% of all bus sales [2,3]. To date, over 430 electric heavy-duty vehicle models have been launched by Chinese manufacturers, more than 170 by North American manufacturers, and over 120 by European manufacturers. Of the 430 models available in China, more than 40% are buses designed for urban public transportation. These developments mark the early stages of a significant global expansion in the electric bus fleet in the years ahead.

Electric buses have a relatively larger energy storage capability and instantaneous power exchange rates compared to the majority of the passenger vehicles. The energy storage

capacity of typical electric buses range from 150 kWh to 350 kWh, while extended range buses can reach beyond 800 kWh [4]. Depot charging power is usually 50 kW to 150 kW, while fast charging powers during operation are in the range of 150 kW to 600 kW. Compared to passenger cars, electric buses, especially the ones used in transport routes operate on fixed schedules with clearly defined departure times, trip routes and depot parking periods make their operation relatively more predictable and offering promising charging coordination options.

Batteries are becoming one of the major assets in grid operation, because of increasing fast flexibility reserve needs due to large penetration of intermittent renewables into the electricity grids worldwide. As of the end of 2024, the battery installations reached a total capacity exceeding 10 GW, surpassing coal and gas in regulation and responsive reserve services since 2022 in the area of Electric Reliability Council of Texas (ERCOT) in the US [5]. Hundreds of MWs of batteries have been effectively utilized in several frequency disturbance events, due to sudden loss of above 1 GW supply, took place in the UK in the recent years [6,7].

V2G has increasingly been integrated into daily lives of EV users recently, with pioneering developments such as the mass-market V2G tariff launched by Octopus Energy in UK, providing free charging if autonomous V2G operation is enabled [8], V2G charger certification activities by The Clean Energy Council (CEC) in Australia [9], bidirectional chargers for home use developed by Wallbox [10] and Emporia [11]. Apart from V2G compatible chargers available for passenger cars, V2G compatible electric bus chargers by Nuvve are in use in the US [12].

Public transport bus fleets are massive in size, such as over 6000 buses operating in Istanbul for more than 800 routes, used by over 2.5 million passengers everyday [13]. In the majority of the time tables (except airport routes, night routes and a few other), the first trip of the day starts later than 06.00 and the last trip of the day usually ends before 23.59. The remaining time (00:00 to 05:59) for most of the buses is devoted to idle parking and waiting at depots, which in case of electrification of bus fleets will also be used for the main charging session of the day for each bus. Considering typical 2 to 4 hours of full charging times for the electric buses on roads today, there still exists 2 or more hours of overnight idle waiting time at depot, satisfying full (or any desired lower) SoC by the beginning of the first trip of the day. The possible overall demand profiles, charging costs and electricity consumption-related emissions of near future

electrified bus fleets is considerable, requiring effective charging coordination strategies. Additionally, bidirectional charging infrastructures and long idle parking-waiting times even after enough energy is transferred to the batteries can allow V2G services provision and additional charging-discharging activities, still satisfying the targeted SoC by the time of first departure.

This study investigates the overnight frequency regulation service provision potential of electric bus public transport fleets. Year-wide system frequency data from 2024 for the UK is analyzed, highlighting the overnight regulation needs (in terms of event occurrence, type and duration) that electric buses can serve for. Taking into account the technical characteristics of commonly preferred electric bus brand and models, along with real routes with characteristic differences.

The second section provides a literature review, while the third section presents the methodology. Section 4 presents the case study with results and the last section concludes the paper.

2. State of the Art

There has been increasing number of studies on V2G, frequency regulation and electric buses (together or separately) in the literature.

Zhang, Liu and Wen (2022) investigated charging scheduling of an electric bus fleet considering V2G services, without discussing in details the occurrence and duration characteristics of specific V2G services [14]. [15] presents an aggregated model for passenger EVs to explore the value of V2G services that can be provided to Transmission System Operator (TSO). Another study [16] focused on frequency control ancillary service provision by shared EVs. [17] aims to maximize individual vehicle owner's profits from selling primary frequency regulation to the grid. [18] investigates reliable and cost-effective charging of electric bus fleets.

The past relevant studies have not explored year-long V2G characteristics, particularly for specific time windows during the day, while this paper analyzes real system frequency data of the UK grid for the whole year of 2024 to highlight the frequency regulation event characteristics that coincide with overnight charging of electric bus public transport fleets.

3. Methodology

This section explains the analysis approach of the system frequency data to quantify frequency regulation needs, electric bus public transport daily mobility activities and overnight charging flexibility potential derivation steps.

3.1. Identification of Frequency Regulation Events Using System Frequency Data

Typical frequency operation range across Europe is 50 Hz. ± 0.05 Hz deviation limits are used as a performance quality target for frequency to stay within for 99.5% of the time by European Network of Transmission System Operators for Europe (ENTSO-E), while ± 0.2 Hz deviation limits are used for the regulation capacity to be fully delivered, and system operation inside tolerance limits of 49 Hz to 51 Hz is allowed [19].

System frequency values are usually available as open datasets, such as in the Data Portal of National Energy System Operator of the UK [20]. The data with 1 second resolution is available for all the past months of the current and past years for

online exploration on a browser-embedded dashboard, and for offline analysis after being downloaded as a .csv file.

After monthly system frequency data with 1 second resolution for all the months of 2024 is downloaded as .csv files, the files are merged into a single file to run data analysis covering the whole year. In the next step, the record with system frequency value below 49.8 Hz or above 50.2 Hz are filtered and listed as the times in which regulation service could be requested from flexibility providers. After that, the filtered records with consecutive time steps are considered as continuing single events, while discrete records that belong to non-consecutive time labels are considered as separate events, listing regulation up and regulation down requiring events for the whole year.

This stage is followed by the analysis of monthly event occurrences that coincides with the usual depot parking periods of public transport buses (00:00 to 06:00 time window), as shown in Fig 1. The highest number of frequency regulation requiring events are faced in October (65 events), while January was the month with least number of events (7). Throughout the year, there were only 287 days (78.63% of the year) without any events, while regulation service was required for at least one time in each of the remaining 78 days (21.37% of the year).

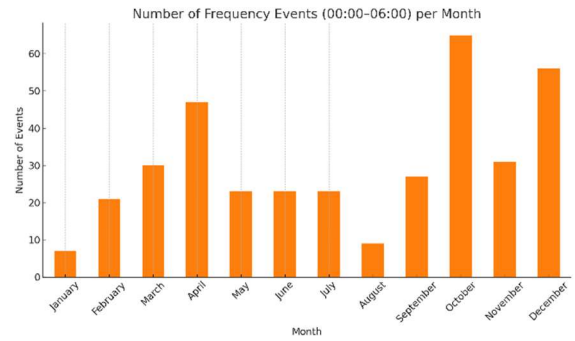


Fig. 1. Monthly Overall Occurrence Rates of Events.

After that event type dominance across the year is analyzed based on the number of occurrences per month as shown in Figure 2. January, March, August and November have dominance of regulation up events, while April, June, July and September witnessed more dominant down events, and in the other months there were similar occurrence rates of up and down events.

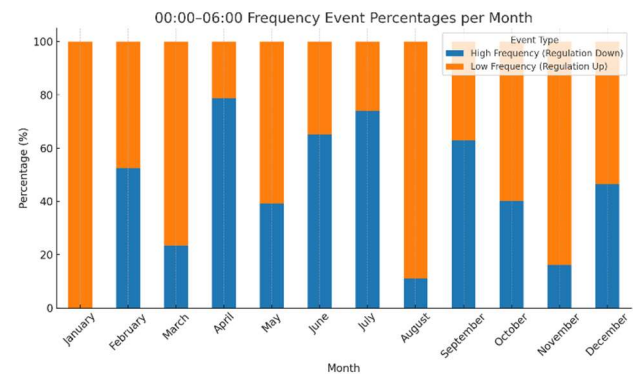


Fig. 2. Monthly Occurrence Percentage of Regulation Down vs Regulation Up Events.

At the next stage overall duration of overnight events are given in seconds, along with overall event durations at any time of the day and the percentage of first in the latter (Table 1).

Table 1. Overall events and overnight events.

Month	Overall Event Durations (s)	Overnight Event Durations (s)	Percentage (%)
January	2109	75	3.56
February	2467	230	9.32
March	3475	627	18.04
April	3539	895	25.29
May	1814	328	18.08
June	2938	458	16.69
July	1378	205	14.88
August	2431	292	12.01
September	2582	318	12.32
October	6077	1044	17.18
November	4774	582	12.19
December	4653	844	18.14

After that, considering individual events, a general histogram of event durations is provided in Figure 3. While 71.82% of the events had a duration less than 10 s, there had been rare events that last up to 5 minutes.

Another histogram is plotted to observe the occurrence of events per day (Figure 4). Around 70% of the days 4 or less events occurred, while there had been a day observed 35 events.

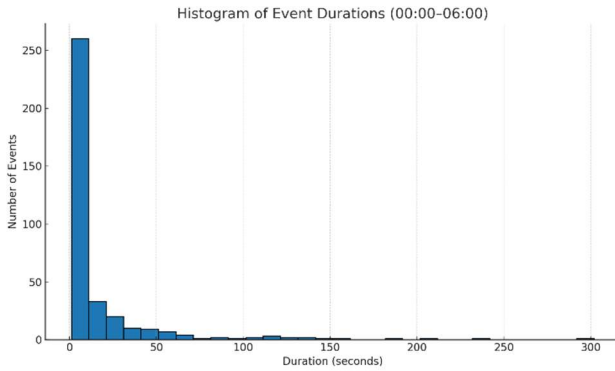


Fig. 3. Histogram of individual event durations.

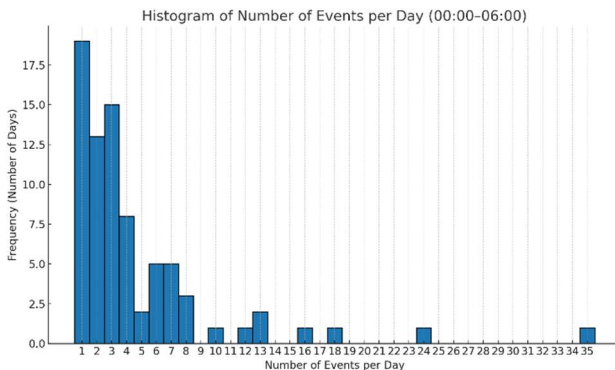


Fig. 4. Histogram of events per day.

3.2. Electric Bus and Route Characteristics

Considering the technical characteristics of widely preferred brand and models in Europe, an energy storage capacity of 320 kWh is selected. Considering electric bus chargers with V2G capability, available in the field applications, a charging/discharging power of 52 kW is selected [12].

Considering the routes in Istanbul, one of the shortest routes is Y1 line with 1.4 km, while one of the longest is 500T with 77 km, majority of the routes range between 10 to 30 km. Energy consumption of electric buses range between 1.5 to 2.5 kWh/km. This requires, 4 min to 3.7 hours of charging to cover for the energy spent in one direction trip. Considering time tables for all the trips of typical days, 5 to 12 buses are needed to successfully conduct every trip throughout the day.

The buses are assumed to arrive the depot after the last trip of the day with an SoC of 20%, an amount that allows minimum one two-way trip couple to prevent any service interruption due to empty battery on the road.

4. Case Study

Considering the buses serve for the earliest starting (usually 06:00) and latest ending (usually 23:59) routes, their depot charging and parking time is limited to 6 hours, which could be longer for some other routes.

The minimum event duration and occurrence are 1 second and 1 event per day respectively. Maximum event duration and occurrence is 302 seconds from 35 events. A fleet of 5 buses and a fleet of 12 buses are considered to be taking part in frequency regulation fleet.

In case of a 1 s single event on a day, 0.014 kWh of energy can be exchanged, relating to less than 1% of the battery's SoC, reducing mileage by around 1 km, while in overall 302 s long 35 events on a day can transfer 4.36 kWh from the battery, around 1.36% of its SoC, accounting for 2.9 km reduction in mileage. A 5-bus fleet can provide 250 kW, while a 12-bus fleet can provide 600 kW.

5. Conclusion

This study explores the potential of electric bus fleets in providing frequency regulation services during their overnight charging and idle times at depots. The analysis, based on the 2024 system frequency data of the UK's National Energy System Operator (NESO), highlights the frequency regulation event characteristics, revealing the temporal patterns and event durations that coincide with the buses' idle periods. The results indicate that electric buses, with their significant energy storage capacities and predictable operational schedules, are well-positioned to contribute to frequency regulation services, particularly during off-peak hours when they are typically parked at depots.

The findings show that frequency regulation events are most prevalent during the autumn months, with varying patterns of regulation up and down events throughout the year. A substantial proportion of these events occur overnight, making electric bus fleets an ideal resource for providing these services without affecting their primary transportation duties. Additionally, the study provides insights into the energy exchange capabilities of electric buses, indicating that even short-duration events, as brief as a few seconds, can contribute to grid stability with minimal impact on the bus fleet's operational range.

The case study further demonstrates that a fleet of 5 to 12 electric buses can provide up to 600 kW of regulation capacity, which could significantly support grid balancing efforts, especially as renewable energy sources continue to expand. This demonstrates the potential of electric bus fleets to participate in grid services underscores the importance of integrating Vehicle-to-Grid (V2G) technologies into public transport infrastructure, ultimately contributing to a more resilient and sustainable energy system.

In conclusion, the integration of electric bus fleets into frequency regulation markets offers a valuable opportunity for both the transport sector and the energy grid. This study provides a foundation for further exploration into the optimization of V2G services, encouraging the adoption of electric buses as versatile assets capable of supporting the transition to a low-carbon energy system while enhancing the operational efficiency of public transport.

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